

Large Deformation Isotropic Elasticity - On the Correlation of Theory and Experiment for Incompressible Rubberlike Solids

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Large deformation isotropic elasticity – on the correlation of theory and experiment for incompressible rubberlike solids

BY R. W. OGDEN

*School of Mathematics and Physics,
University of East Anglia, Norwich*

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Many attempts have been made to reproduce theoretically the stress–strain curves obtained from experiments on the isothermal deformation of highly elastic ‘rubberlike’ materials. The existence of a strain-energy function has usually been postulated, and the simplifications appropriate to the assumptions of isotropy and incompressibility have been exploited. However, the usual practice of writing the strain energy as a function of two independent strain invariants has, in general, the effect of complicating the associated mathematical analysis (this is particularly evident in relation to the calculation of instantaneous moduli of elasticity) and, consequently, the basic elegance and simplicity of isotropic elasticity is sacrificed. Furthermore, recently proposed special forms of the strain-energy function are rather complicated functions of two invariants.

The purpose of this paper is, while making full use of the inherent simplicity of isotropic elasticity, to construct a strain-energy function which: (i) provides an adequate representation of the mechanical response of rubberlike solids, and (ii) is simple enough to be amenable to mathematical analysis. A strain-energy function which is a *linear* combination of strain invariants defined by $\phi(\alpha) = (\alpha_1^2 + \alpha_2^2 + \alpha_3^2 - 3)/\alpha$ is proposed; and the principal stretches α_1 , α_2 and α_3 are used as independent variables subject to the incompressibility constraint $\alpha_1 \alpha_2 \alpha_3 = 1$. Principal axes techniques are used where appropriate. An excellent agreement between this theory and the experimental data from simple tension, pure shear and equibiaxial tension tests is demonstrated.

It is also shown that the present theory has certain repercussions in respect of the constitutive inequality proposed by Hill (1968*a*, 1970*b*).

1. INTRODUCTION

This paper is concerned with the mathematical representation of the isothermal mechanical behaviour of materials which may broadly be classified as ‘rubberlike’. Specifically the term rubberlike will be used to describe those solids which, in common with natural rubber, are highly elastic. At a fixed temperature such materials are very nearly perfectly elastic for quite large quasi-static deformations from the undistorted state; in other words, hysteresis effects are small. Thus, from a theoretical standpoint, it is convenient to regard these materials as perfectly elastic and to postulate the existence of a strain- (stored-) energy function† by means of which their mechanical properties may be characterized. It is supposed throughout that the temperature remains constant during deformation.

The strain-energy function depends only on the terminal states of strain and in no

† Elastic materials which possess a strain-energy function are called Green-elastic or hyperelastic materials. More general elastic materials (Cauchy-elastic), for which there is no strain-energy function, may be defined but this is neither necessary nor desirable for the purposes of this paper.

way on the history of straining. Relative to a given reference configuration the state of strain at each material element is completely specified by the three principal stretches, denoted by a_1 , a_2 and a_3 , measured from that configuration, together with the directions of the principal axes of strain.†

Experience suggests that rubberlike solids are isotropic relative to the undistorted state, and the assumption of isotropy has therefore played an important role in simplifying the associated mathematical analysis. Furthermore, rubberlike solids are substantially incompressible except under certain extreme conditions, so the additional simplifying assumption of incompressibility has usually been adopted.

For an *isotropic* elastic solid the strain-energy function, denoted by ϕ , depends on the strain only through the principal stretches a_1 , a_2 and a_3 relative to some ‘ground state’. Moreover, $\phi(a_1, a_2, a_3)$ measured per unit volume in the ground state, must be symmetric with respect to interchange of any two of a_1 , a_2 and a_3 . In this paper we shall assume the existence of an undistorted, stress-free state and this will be taken as the ground state, though any state differing from this by a pure dilatation (for compressible materials) is also a possible ground state. All strains are measured from the chosen ground state. When the material is incompressible the constraint

$$a_1 a_2 a_3 = 1 \quad (1)$$

must be satisfied identically throughout the material. Here attention will be confined to incompressible materials, but the present work will be extended to account for the effects of compressibility in a subsequent paper.

The basic problem is to construct strain-energy functions consistent with the observed behaviour of rubberlike solids when deformed isothermally. During the past 20 to 30 years many attempts have been made to determine the form of the strain-energy function, and it was usually thought expedient to use certain stretch or strain invariants as independent variables in preference to the principal stretches. Such invariants are themselves symmetric in a_1 , a_2 and a_3 . Those in most common use are defined by

$$I_1 = a_1^2 + a_2^2 + a_3^2, \quad I_2 = a_2^2 a_3^2 + a_3^2 a_1^2 + a_1^2 a_2^2 \equiv a_1^{-2} + a_2^{-2} + a_3^{-2} \quad (2)$$

for incompressible materials so that the strain energy may be specified as a function of I_1 and I_2 . However, such choices of independent variable in general needlessly complicate the associated mathematical analysis. This is particularly evident in relation to the calculation of ‘instantaneous’ moduli of elasticity, as pointed out by Hill (1970*a*). Principal axes techniques, which were developed in the present context by Hill (1970*a, b*), obviate the need for any special choice of invariants and, moreover, by use of such techniques, the basic elegance and simplicity of isotropic elasticity is underlined.

Our objective is to reassess the situation regarding the relationship between theory and experiment for rubberlike solids in the light of the remarks made in the

† By principal axes of strain we mean here the principal axes of the Lagrangian (or spatial) strain ellipsoid as distinct from those of the Eulerian (or material) strain ellipsoid.

preceding paragraph. With this in mind, previously proposed forms of the strain-energy function are discussed critically in §2 along with the relevant experimental work. Next, in §3, we propose a strain-energy function which is a *linear* combination of strain invariants defined by $\phi(\alpha) = (a_1^\alpha + a_2^\alpha + a_3^\alpha - 3)/\alpha$, where α is a real number. We write†

$$\phi = \sum_r \mu_r \phi(\alpha_r)$$

and seek values of the constants α_r and μ_r which provide a good fit to the experimental data. The principal stretches are used as independent variables subject to the incompressibility constraint (1) and the fullest simplicity, appropriate to isotropy, is retained throughout the analysis. In §4 the strain-energy function is related to the experimental results of Treloar (1944) on vulcanized natural rubber, and an excellent correlation is demonstrated by use of three (α_r, μ_r) pairs. The problem of inflation of a spherical balloon is then considered in §5 on the basis of the present theory. The repercussions of the theory in respect of a certain constitutive inequality due to Hill (1968*a*, 1970*b*) are discussed in §6, and, in particular, it is shown that the conditions $\mu_r \alpha_r > 0$ (each r) are sufficient for this inequality to be satisfied. Some kinematical results which are used in §6 are derived in the appendix.

2. PREVIOUS THEORIES

Treloar (1943), on the basis of Gaussian statistics and molecular network theory, constructed the so-called neo-Hookean form of the strain energy, namely

$$\phi = \frac{1}{2}\mu(I_1 - 3), \quad (3)$$

where μ is the shear (or rigidity) modulus in the ground state and I_1 is defined by equation (2)₁. The first systematic series of experiments on the mechanical properties of rubberlike solids was carried out by Treloar (1944) who studied the response of vulcanized natural rubber in simple tension, pure shear and uniform equibiaxial tension. The primary objective of these experiments was to determine the range of validity of the neo-Hookean strain energy (3). This simple form provides an adequate first approximation to the behaviour of rubberlike solids and is generally regarded as a valid prototype for this class of materials. It has the advantage of being easy to treat mathematically so that solutions to many problems have been obtained by its use.

The specimens used by Treloar in the three distinct experimental stress systems were obtained from a single sample of vulcanized rubber in the form of a thin sheet, thus ensuring consistency when correlating theory and experiment. The range of strain achieved in these experiments was large (about 750 % extension in the simple tension test) and the results are representative of those for other rubberlike solids. Consequently Treloar's data have been used as the basis for comparison with theory

† The stimulus for an investigation of such functions comes from the work of Dr R. Hill, F.R.S. He made a rough comparison of the single-term formula with data at large strains and obtained encouraging results.

by several authors (e.g. Carmichael & Holdaway 1961; Klingbeil & Shield 1964; Hart-Smith 1966; Hart-Smith & Crisp 1967; Alexander 1968). A further advantage of this particular rubber is that the effects of hysteresis and crystallinity are small.

The strain-energy function defined by

$$\phi = \mu(a_1 + a_2 + a_3 - 3) \quad (4)$$

was proposed by Varga (1966) specifically as a first approximation to the behaviour of rubberlike solids. He has demonstrated that the range of validity of (4) is comparable with that of (3). It should be pointed out that (4) is not expressible as a *simple* function of the invariants I_1 and I_2 .

The general mathematical theory of large deformation isotropic elasticity was developed by Rivlin (1948*a-d*, 1949*a-c*) in a series of papers in which the invariants I_1 and I_2 were used (for incompressible materials) as independent variables. Subsequently, Rivlin & Saunders (1951) based a number of experiments on this theory in an attempt to determine the form of the strain energy as a function of I_1 and I_2 , $\phi(I_1, I_2)$. They did not achieve such large strains as did Treloar (in fact only about 370 % extension in simple tension) because they wished to eliminate the effects of crystallization at higher strains. However, as pointed out by Treloar (1958, pp. 18, 19), the effect of crystallization is only secondary as regards elastic properties so that this restriction was unnecessary. In addition to the three tests mentioned above Rivlin & Saunders performed, amongst others, the following tests on solid circular rubber cylinders: (*a*) simple torsion, and (*b*) torsion superposed upon simple elongation. These deformations will be discussed elsewhere.

The theory employed by Rivlin & Saunders involves the calculation of $\partial\phi/\partial I_1$ and $\partial\phi/\partial I_2$ from the experimental measurements. However, for certain stress systems, the formulae used accentuate the experimental errors so that values of $\partial\phi/\partial I_1$ and $\partial\phi/\partial I_2$ cannot be determined accurately enough for some ranges of values of I_1 and I_2 . This demonstrates one disadvantage of using the theory based upon the invariants I_1 and I_2 . Such disadvantages do not occur in the theory to be developed in the later sections of this paper. The results of their experiments led Rivlin & Saunders to suggest a strain-energy function of the form

$$\phi = \frac{1}{2}\mu(I_1 - 3) + f(I_2 - 3) \quad (5)$$

over the range of strain they considered. They did not specify the function f , but (5) has been the subject of some more recent investigations (e.g. Gent & Thomas 1958). It is known (see Hart-Smith 1966, §1) that (5) does not give a good representation of Treloar's data so that applications of the Rivlin-Saunders theory are limited.

It was pointed out by Mooney (1940) and Rivlin (1949*c*) that the strain-energy function can be expanded as an infinite series in terms of I_1 and I_2 . Thus

$$\phi = \sum_{m,n=0}^{\infty} C_{mn}(I_1 - 3)^m (I_2 - 3)^n \quad (C_{00} = 0), \quad (6)$$

where the C_{mn} 's are constants. The neo-Hookean form (3) is a special case of this, as also is the Mooney (or Mooney–Rivlin) form

$$\phi = C_{10}(I_1 - 3) + C_{01}(I_2 - 3), \quad (7)$$

which is *linear* in the invariants I_1 and I_2 . With a suitable choice of the constants C_{01} and C_{10} the Mooney form of strain-energy function gives a marginally better fit to the experimental data than the neo-Hookean form. Equation (7) was derived by Mooney (1940) as the most general form of strain-energy function admitting a linear relationship between stress and strain in simple shear.

Higher order specializations of (6) have been considered by Isihara, Hashitsume & Tatibana (1951), Biderman (1958) and Ko & Blatz (1963), among others. However, correlation between these forms and experimental data is not significantly better than for the Mooney form, and, moreover, analysis of these strain-energy functions becomes unwieldy as soon as the linearity of (7) is lost. Clearly, little progress is likely to be made by pursuing an approach based on the general expansion (6).

In attempting to fit Treloar's data Hart-Smith (1966) constructed the strain-energy function

$$\phi = \mu_1 \int \exp\{\nu_1(I_1 - 3)^2\} dI_1 + \mu_2 \ln(I_2/3). \quad (8)$$

This gives good correlation with the data for small and moderate strains, and the resulting stress–strain curves have the correct trend at high strains. However, Alexander (1968) found that (8) was not suitable for the synthetic rubber *neoprene*† at moderate strains. On the basis of his own experiments on neoprene Alexander constructed the modified form

$$\phi = \mu_1 \int \exp\{\nu_1(I_1 - 3)^2\} dI_1 + \mu_2 \ln\left(\frac{I_2 - 3 + \nu_2}{\nu_2}\right) + \mu_3(I_2 - 3), \quad (9)$$

which gives a close correspondence with the experimental data. In (8) and (9) the parameters μ_r ($r = 1, 2, 3$) and ν_s ($s = 1, 2$) are constants whose values are selected by comparing the predictions of (8) and (9) with the relevant data.

Any mathematical analysis based on the general expansion (6) or on complicated forms such as (8) and (9) tends to be cumbersome, particularly in relation to problems in which the principal axes of strain vary through the material. It is therefore apparent that an adequate correlation between theory and experiment for a wide range of strains has been achieved only at the expense of mathematical simplicity. The same is true of the theory of Carmichael & Holdaway (1961) who, departing from the practice of writing the strain energy as a function of I_1 and I_2 , assumed a separable form, namely $\phi = \Phi(a_1^2) + \Phi(a_2^2) + \Phi(a_3^2) - 3\Phi(1)$. An equivalent separation has been proposed more recently by Valanis & Landel (1967) and utilized by Obata, Kawabata & Kawai (1970) and Peng & Tschoegl (1971) *inter alia*.

In the light of the above discussion it is our objective to develop a strain-energy function which is free from the shortcomings mentioned. Briefly, we are seeking a

† This material is used in the construction of high altitude meteorological balloons.

strain-energy function which (i) provides an accurate representation of the mechanical response of rubberlike materials for large ranges of deformation, and (ii) is simple enough to be amenable to mathematical analysis.

3. THE PROPOSED STRAIN-ENERGY FUNCTION

We are concerned with elastic solids which possess a strain-energy function and which are isotropic relative to the stress-free ground state. We assume that the solid is incompressible so that the principal stretches are connected via equation (1). In these circumstances the principal Cauchy stresses σ_1 , σ_2 and σ_3 are written simply as

$$\sigma_i = a_i \frac{\partial \phi}{\partial a_i} - p \quad (i = 1, 2, 3), \quad (10)$$

where ϕ is the strain-energy function, and p is an arbitrary hydrostatic pressure introduced because of the incompressibility constraint. Because of the isotropy the principal axes of the Cauchy stress tensor coincide with those of the Eulerian strain ellipsoid, i.e. the finally orthogonal triad of embedded line elements which were orthogonal and equal in the ground state. The principal axes of the right stretch tensor are coincident with the axes of the Lagrangian strain ellipsoid which are in the reference directions of the embedded triad of line elements that currently define the axes of the Eulerian strain ellipsoid.

We consider the class of strain measures, coaxial with the Lagrangian strain ellipsoid, having principal values e_i ($i = 1, 2, 3$) given by

$$e_i = \begin{cases} (a_i^\alpha - 1)/\alpha & (\alpha \neq 0), \\ \ln a_i & (\alpha = 0), \end{cases} \quad (11)$$

where α is a real number, positive or negative. More general strain measures may be defined (cf. Hill 1968*a*), but they are not required here. The first invariants (i.e. the traces of the corresponding matrices) of these strain measures are expressible as

$$\phi(\alpha) = \begin{cases} (a_1^\alpha + a_2^\alpha + a_3^\alpha - 3)/\alpha & (\alpha \neq 0), \\ \ln a_1 a_2 a_3 \equiv 0 & (\alpha = 0), \end{cases} \quad (12)$$

each of which is symmetric in a_1 , a_2 and a_3 . Henceforth, we exclude $\alpha = 0$ from consideration. The two invariants $\phi(\alpha)$ and $\phi(-\alpha)$ may be regarded as a pair of independent invariants of the strain measure corresponding to α (or $-\alpha$) and, as α varies through both positive and negative values, $\phi(\alpha)$ covers a complete range of possible pairs of independent invariants. The invariants I_1 and I_2 , defined by equations (2), are related to the class (12) by $\phi(2) = \frac{1}{2}(I_1 - 3)$ and $\phi(-2) = -\frac{1}{2}(I_2 - 3)$. It is not our intention to use any such pair nor, indeed, any other possible pair of invariants as independent variables.

We now propose a strain-energy function which is a *linear* combination of the invariants $\phi(\alpha)$ and write†

$$\phi = \mu_r \phi(\alpha_r), \quad (13)$$

where the μ_r 's are constants and summation over r is implied. The number of terms included in the summation is to be determined by comparison of this theory with experiment in §4. One possible combination is the set of terms corresponding to integral values of the α_r 's, in which case we may write

$$\phi = \sum_{n=-\infty}^{\infty} \mu_n \phi(n),$$

where n is an integer. We shall not use this form but, in preference, the more general form (13), allowing the α_r 's to take both integral and non-integral values. As will be shown in §4 this facilitates the correlation with experimental data and permits a very good fit to the data with a small number of terms. If only integral values of the α_r 's were considered more terms would be needed to achieve as close a fit.

According to equation (10) the principal Cauchy stresses resulting from (13) are simply

$$\sigma_i = \mu_r \alpha_i^{\alpha_r} - p \quad (i = 1, 2, 3), \quad (14)$$

with summation over r (this summation being implied henceforth). It is interesting to note that on specializing equations (14) to the classical linear theory of isotropic elasticity the conventional shear modulus μ is related to the parameters μ_r and α_r by

$$2\mu = \mu_r \alpha_r \quad (\text{summed over } r).$$

In order to prepare for comparison of the present theory with experiment the equations to which (14) reduce in simple tension, pure shear and equibiaxial tension will be written down explicitly. These three stress systems correspond to pure homogeneous deformations, i.e. they are homogeneous deformations for which the principal axes of strain do not vary in direction relative to the material either with position in the body or with strain. Simple shear is a pure shear together with a rotation of these principal axes; for contrast, this will also be discussed. The problems of simple torsion (locally equivalent to simple shear) and torsion superposed upon axial extension of rubber cylinders are also of interest; these problems, among others, will be discussed elsewhere.

(a) Simple tension

Let $a_1 = a$ be the stretch ratio in the direction of elongation and $\sigma_1 = \sigma$ the corresponding principal Cauchy stress. The other two principal stresses are zero

† The most general form of isotropic expansion of ϕ in terms of a_1 , a_2 and a_3 involves expressions of the form

$$a_1^\alpha (a_2^\beta + a_3^\beta) + a_2^\alpha (a_3^\beta + a_1^\beta) + a_3^\alpha (a_1^\beta + a_2^\beta), \quad (*)$$

but such generality is not needed here. All possible invariants of the strain measures (11) are expressible as linear combinations of expressions of the form (*).

since no lateral forces are applied; so, by virtue of the incompressibility constraint, $a_2 = a_3 = a^{-\frac{1}{2}}$. The independent stress-strain relations are, from equations (14),

$$\sigma = \mu_r a^{\alpha_r} - p \quad \text{and} \quad 0 = \mu_r a^{-\frac{1}{2}\alpha_r} - p.$$

Elimination of p yields $\sigma = \mu_r(\alpha^{\alpha_r} - a^{-\frac{1}{2}\alpha_r})$.

Measurements in such tension tests are usually made in terms of the force per unit unstrained area of cross-section. This force, denoted by f , is given by

$$f = \sigma a^{-1} = \mu_r[a^{\alpha_r-1} - a^{-(1+\frac{1}{2}\alpha_r)}]. \quad (15)$$

In simple tension $a > 1$, while the same equation applies for unidirectional compression (equivalent to equibiaxial tension) when $a < 1$.

(b) *Pure shear*

In pure shear one of the principal extension ratios is held fixed, $a_2 = 1$ say. Setting $a_1 = a$ and $a_3 = a^{-1}$ the stress-strain relations (14) reduce to

$$\sigma_1 = \mu_r a^{\alpha_r} - p, \quad \sigma_2 = \sum_r \mu_r - p, \quad 0 = \mu_r a^{-\alpha_r} - p.$$

We are only concerned with the direction of extension here so the force f per unit undeformed area of cross-section in this direction is given by

$$f = \sigma_1 a^{-1} = \mu_r[a^{\alpha_r-1} - a^{-(1+\alpha_r)}], \quad (16)$$

on elimination of p , as in (a).

(c) *Equibiaxial tension*

In equibiaxial tension two of the principal stresses are equal, $\sigma_2 = \sigma_3 = \sigma$ say, while the third is zero. Correspondingly the stretches are written $a_2 = a_3 = a$ and $a_1 = a^{-2}$. Elimination of p from the two remaining independent equations (14) yields

$$\sigma = \mu_r(a^{\alpha_r} - a^{-2\alpha_r}).$$

Consider a thin sheet of rubber with initial thickness h and let T be the force (surface tension) acting on a section of unit length cut at right angles to the plane of the sheet. In the deformed state the thickness of such a section is $ha_1 = ha^{-2}$. Thus $T = \sigma ha^{-2}$ and

$$Th^{-1} = \mu_r[a^{\alpha_r-2} - a^{-(2+2\alpha_r)}].$$

In order to retain uniformity with equations (15) and (16), and to facilitate comparison with experiment, we write this in the alternative form

$$f \equiv Tah^{-1} = \mu_r[a^{\alpha_r-1} - a^{-(1+2\alpha_r)}], \quad (17)$$

so that the leading exponent on the right-hand side of each of equations (15), (16) and (17) is $\alpha_r - 1$, and f has the same physical interpretation as in (15) and (16).

Equibiaxial tension is most effectively achieved by inflation of a circular membrane into spheroidal shape. The stress field in the region of the pole of the spheroid

can be regarded as corresponding to equibiaxial tension (by superposition of an appropriate hydrostatic pressure) provided the skin thickness h is very much less than the radius of curvature r at the pole. The pressure of inflation P is then related to the surface tension T in the skin via the approximate formula

$$P = 2Tr^{-1} \quad (18)$$

to the first order in hr^{-1} . By means of equation (18) T can be calculated from measured values of r and P , and comparison made with the predictions of equation (17).

(d) Simple shear

Simple shear is an isochoric plane deformation, so we may take $a_1 = a$, $a_2 = a^{-1}$ and $a_3 = 1$. The two-dimensional deformation gradient, denoted by A , has the matrix representation

$$A = \begin{pmatrix} 1 & \gamma \\ 0 & 1 \end{pmatrix},$$

where γ is the amount of shear; let A' denote the transpose of A . The left Cauchy-Green matrix AA' defines the axes of the Eulerian strain ellipsoid which we assume to have orientation θ relative to background rectangular Cartesian axes (x, y) . A short calculation leads to

$$a - a^{-1} = \gamma, \quad a + a^{-1} = \sqrt{4 + \gamma^2}$$

and

$$\tan 2\theta = 2\gamma^{-1}.$$

The relevant principal stresses are

$$\sigma_1 = \mu_r a^{\alpha_r} - p, \quad \sigma_2 = \mu_r a^{-\alpha_r} - p$$

and the shear stress σ_{xy} is therefore given by

$$\sigma_{xy} = \frac{1}{2}(\sigma_1 - \sigma_2) \sin 2\theta = \mu_r \frac{a^{\alpha_r} - a^{-\alpha_r}}{a + a^{-1}}. \quad (19)$$

When $\alpha_r = 2$ or -2 this reduces to a linear relationship between shear stress and shear strain, as is well known. It is worth noting that the normal stresses are

$$\sigma_{xx} = \mu_r \frac{a^{\alpha_r+1} + a^{-(\alpha_r+1)}}{a + a^{-1}} - p, \quad \sigma_{yy} = \mu_r \frac{a^{\alpha_r-1} + a^{-(\alpha_r-1)}}{a + a^{-1}} - p.$$

4. CORRELATION WITH EXPERIMENT

The experimental data of Treloar (1944) in simple tension, pure shear and equibiaxial tension is plotted in figures 1(a), (b) and (c) respectively according to the scheme of equations (15), (16) and (17). The units† of f are kg cm^{-2} . For comparison with the data the curves resulting from the neo-Hookean and Varga strain-energy functions are plotted in the same figures with appropriately chosen values of the parameter μ (such that $\mu\alpha = 8.0 \text{ kg cm}^{-2}$ in each case). These are two particular cases of the strain-energy function defined by equation (13) and represent the single term specializations of (13) corresponding to $\alpha = 2$ and $\alpha = 1$ respectively.

† It is not convenient to use a non-dimensionalized form of f here.

On inspection of figures 1(a) and 1(b) we see that the two strain-energy functions correspond closely to the data for stretch ratios up to about 1.4. Above this value the experimental points lie between the two curves for values of a up to about 5.5. Above $a \simeq 5.5$ the experimental points in simple tension rise rapidly above the $\alpha = 2$ curve; one would expect the same effect in pure shear but extensions of only about 5.0 were achieved in this deformation. From figure 1(c) we see that the $\alpha = 1$ curve fits the biaxial data up to $a \simeq 1.5$ and the $\alpha = 2$ curve fits the data up to $a \simeq 2.7$. For $a > 1.5$ the $\alpha = 1$ curve drops below the experimental points as also does the $\alpha = 2$ curve for $a > 2.7$.

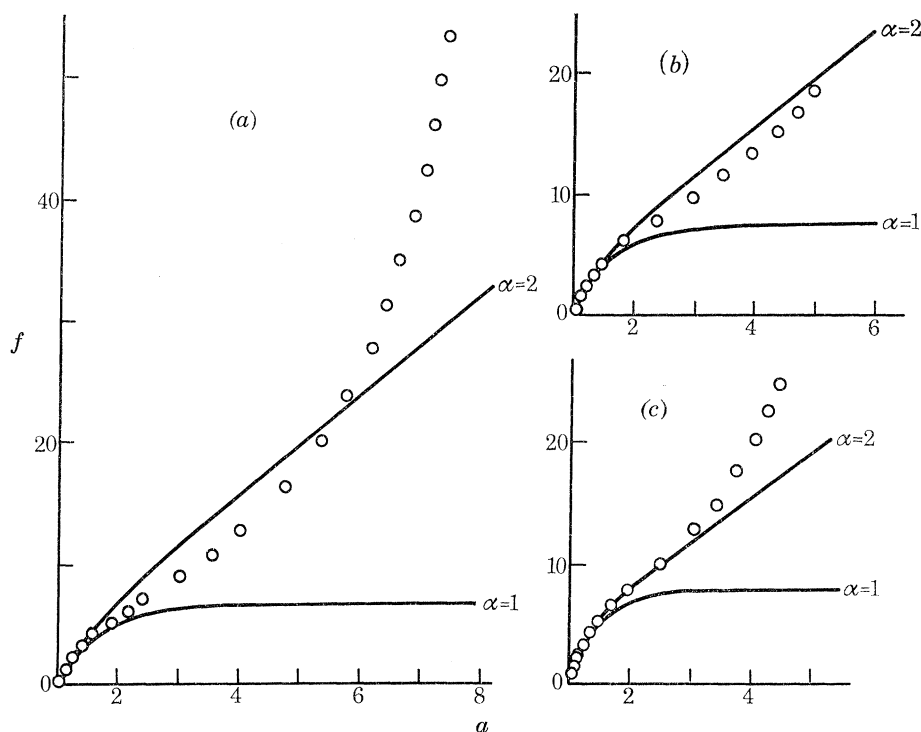


FIGURE 1. (a) Treloar's data in simple tension compared with the neo-Hookean ($\alpha = 2$) and Varga ($\alpha = 1$) theories. (b) Treloar's data in pure shear compared with the neo-Hookean ($\alpha = 2$) and Varga ($\alpha = 1$) theories. (c) Treloar's data in equibiaxial tension compared with the neo-Hookean ($\alpha = 2$) and Varga ($\alpha = 1$) theories.

Our immediate task is to find a value of α which gives a better overall fit to the data in the three stress systems than either of the values of α considered above. Clearly an appropriate value of α lies between 1 and 2. For demonstration purposes the curves corresponding to $\alpha = 1.2$ with $\mu = 6.8 \text{ kg cm}^{-2}$ are drawn in figures 2(a), (b) and (c) compared with Treloar's data. Any value of α between 1.2 and about 1.3 serves equally well. The resulting overall fit is very good up to $a \simeq 2.0$. Increasing the value of α improves the fit to the biaxial data at the expense of that in pure shear and simple tension. Decreasing α has the reverse effect.

Having obtained a good overall fit to the data up to stretches of about 2.0 with the single-term strain-energy function $\phi = \mu_1 \phi(\alpha_1)$, we wish to build on this basis by introducing additional (α, μ) terms in order to achieve a good fit to the full range of considered data. First, however, we note that it is possible (and in certain circumstances it may be convenient) to approximate a given situation by use of a single-term strain-energy function. This can be done by determining an appropriate value of α as follows.

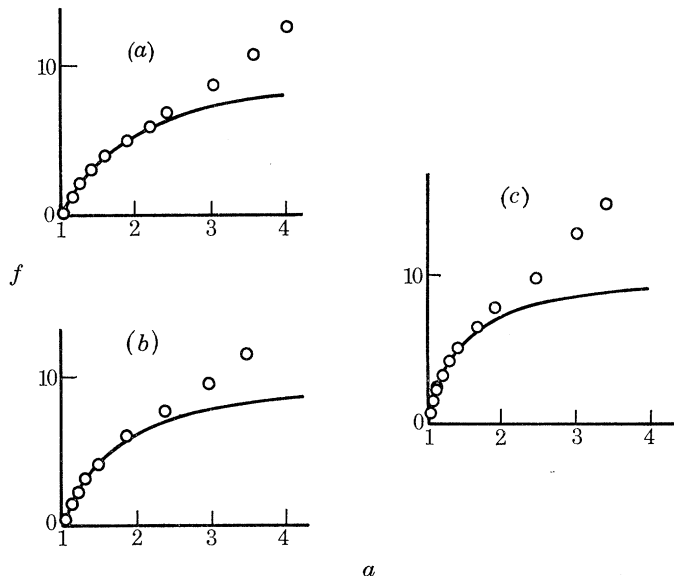


FIGURE 2. Treloar's data compared with the single-term theory $\alpha = 1.2$, $\mu = 6.8$ in (a) simple tension, (b) pure shear and (c) equibiaxial tension.

Since the data we have used are fitted well up to $a \simeq 2.0$ with a single (α, μ) term it is only necessary to consider values of a greater than 2.0. Corresponding to a single (α, μ) term the stress-strain relations in simple tension, pure shear and equibiaxial tension are

$$\sigma = \mu(a^\alpha - a^{-\frac{1}{2}\alpha}), \quad \sigma = \mu(a^\alpha - a^{-\alpha}), \quad \sigma = \mu(a^\alpha - a^{-2\alpha})$$

respectively, where σ is the principal Cauchy stress in the relevant direction. For α sufficiently greater than zero and $a > 2.0$ the dominant term in each case is μa^α (provided $a^{3\alpha/2}$, $a^{2\alpha}$, $a^{3\alpha} \gg 1$ respectively; by an order of magnitude, say). Thus $\ln \sigma \simeq \ln \mu + \alpha \ln a$ and, from a plot of $\ln \sigma$ against $\ln a$, the gradient of any approximately straight line portion can be read off, as can the intercept $\ln \mu$. For example, a value of α of about 5.0 leads to a good fit to the simple tension data for values of a between 6.0 and 7.0.

Thus, we can fit the simple tension data at 'small' and at 'large' strains, in each case with a single (α, μ) term. We now form the strain-energy function consisting of a linear combination of these two terms, (α_1, μ_1) and (α_2, μ_2) , and make an appropriate adjustment in the value of the ratio μ_2/μ_1 . This enables the two terms to be matched

so that a close fit to the data at intermediate strains is also achieved. The value of the ratio μ_2/μ_1 is such that the (α_2, μ_2) term gives no significant contribution to the value of f for values of a up to about 2.0; the fit of the single term (α_1, μ_1) is then disturbed as little as possible in this range. Nevertheless, it is necessary to make slight adjustments in the values of α_1 and μ_1 in order to accommodate the second term satisfactorily. Happily, the same two terms also provide a good fit to the pure shear data. By way of illustration, the curves resulting from a typical two-term strain-energy function $\phi = \mu_1\phi(\alpha_1) + \mu_2\phi(\alpha_2)$ are plotted in figure 3 and compared with the simple tension and pure shear data of Treloar. The values $\alpha_1 = 1.3$, $\alpha_2 = 5.0$, $\mu_1 = 6.6 \text{ kg cm}^{-2}$ and $\mu_2 = 0.012 \text{ kg cm}^{-2}$ have been used and it is seen that the correspondence between theory and experiment is very good up to $a \simeq 7.0$. Within limits these values may be varied without impairing the quality of the fit.

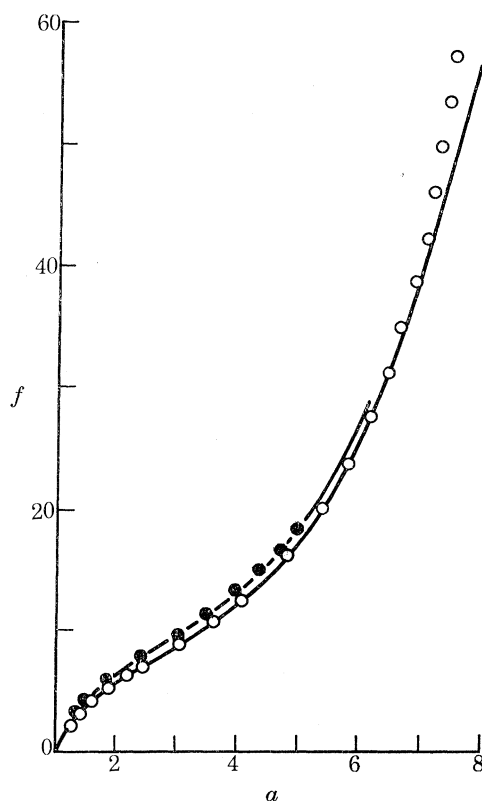


FIGURE 3. Two-term theory compared with Treloar's data in simple tension (○), and pure shear (●).

Unfortunately a two-term strain-energy function is not sufficient to provide a good fit in biaxial tension for values of a greater than about 2.0 so it is necessary to introduce a third term (α_3, μ_3) . It will be noticed on inspection of figures 1 (*a*), (*b*) and (*c*) that the curves of the simple tension and pure shear data are quite close together, whereas that of the biaxial data rises more rapidly and separates away

from the other two as a increases (this can be seen more clearly in figure 4). The relative contributions to f from a single $\phi(\alpha)$ term are

$$a^{\alpha-1} - a^{-(1+\frac{1}{2}\alpha)}, \quad a^{\alpha-1} - a^{-(1+\alpha)}, \quad a^{\alpha-1} - a^{-(1+2\alpha)}$$

respectively in simple tension, pure shear and equibiaxial tension. Clearly α_3 must be negative and less than $-\frac{1}{2}$ for the biaxial curve to separate significantly from the other two as a increases. Correspondingly, μ_3 must be negative and of such an order of magnitude that this third term gives relatively little contribution to the curves for small strains. Again it may be necessary to make slight adjustments in the values of (α_1, μ_1) and also of (α_2, μ_2) . Taking $\alpha_3 = -2.0$ and $\mu_3 = -0.1 \text{ kg cm}^{-2}$ the dominant contributions to the three curves from the (α_3, μ_3) term are in the ratio $1:a:a^3$ for a sufficiently greater than unity. The value of μ_3 ensures that there is little contribution from the first two for the range of strain we are concerned with but a significant contribution from the third.

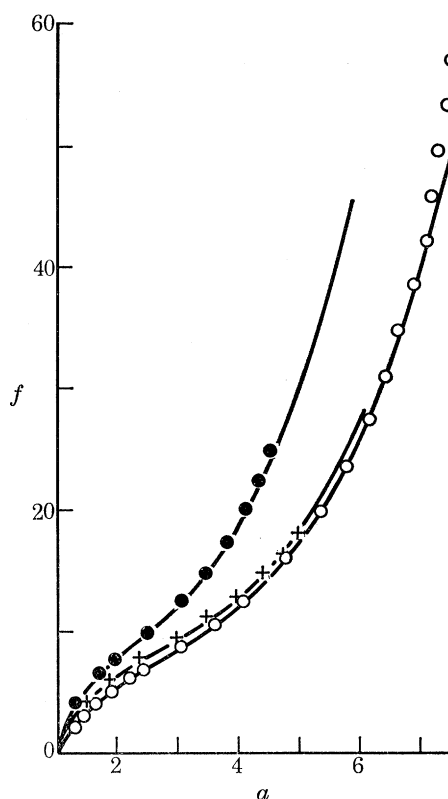


FIGURE 4. Three-term theory compared with Treloar's data in simple tension (○), pure shear (+) and equibiaxial tension (●).

In figure 4 are plotted the curves corresponding to the three-term strain-energy function with values

$$\alpha_1 = 1.3, \quad \alpha_2 = 5.0, \quad \alpha_3 = -2.0;$$

$$\mu_1 = 6.3, \quad \mu_2 = 0.012, \quad \mu_3 = -0.1 \text{ kg cm}^{-2},$$

compared with Treloar's data. The fit is now very good in all three stress systems up to $a \simeq 7.0$. Refinement of the above values of the parameters by use of computer calculations could improve the already good fit, but it is not thought necessary to do this here since the general correspondence between the present strain-energy function and experiment has been adequately demonstrated. Clearly we could also fit the simple tension data more closely for $a > 7.0$ by introducing a fourth term (α_4, μ_4) into the strain-energy function. A value of α_4 of about 10 would be required together with a suitably small value of μ_4 so that this fourth term would give no significant contribution to f for $a < 7.0$. Of course, further small adjustments in the values of the first three pairs of parameters would be necessary in this case. The overall fit of the three-term theory, as shown in figure 4, is very good. But, it is apparent that the simple tension curve constructed from our theory does not rise rapidly enough for very large strains. One would expect this to be the case for the pure shear and equibiaxial tension curves for strains at which data is not presently available. Recent data of Alexander (1971) on the inflation of spherical balloons bears out this conjecture to a certain extent. However, these effects can be accommodated by our strain-energy function if the need arises by either introducing a fourth term into the strain-energy function as mentioned above or modifying the values of the first three pairs of parameters (for example, by increasing $|\alpha_3|$ sufficiently while adjusting the values of the other parameters if necessary).

In relating our strain-energy function to the experimental data we have so far restricted attention to three simple stress configurations. This is partly for convenience, but mainly because there is a lack of comprehensive experimental data for other configurations, particularly in relation to a single rubberlike solid. However, other stress systems are considered by Ogden & Chadwick (1972) in the context of the present theory. Some non-homogeneous deformations, in which the principal axes of strain are not fixed in direction, are treated analytically and the resulting stress distributions obtained in simple form. The consequences are then compared with the limited data available for such deformations though these data are for rubbers different from the one for which our strain-energy function was constructed. The general mechanical characteristics of all these rubbers are the same and, as mentioned in §2, the particular rubber with which we have been concerned is representative of them. The numerical values of the parameters α_r and μ_r found here characterize specifically this one rubber so that different values of the parameters are required for rubbers of differing constitution. Such values will not be noted here but they can be found, if required, by the same procedure as that described in this section. Thus, our strain-energy function is readily adaptable to other rubberlike solids. Analysis of the strain-energy function (13) is straightforward; in fact, it is no more difficult in principle than for a single $\phi(\alpha)$ term because of the linearity. This is demonstrated for some cylindrical problems by Ogden & Chadwick (1972).

5. INFLATION OF A SPHERICAL BALLOON

It is interesting to apply our theory to a particular, though somewhat idealized, problem, namely the investigation of the behaviour of a spherical balloon undergoing inflation. Experience suggests that as a balloon is inflated the applied pressure increases to a maximum as the balloon expands and that further inflation is achieved with a reduced pressure.† Eventually, as one would expect, the pressure reaches a minimum and then increases indefinitely until the balloon bursts. These effects have been adequately demonstrated by Alexander (1971), who performed experiments on balloons made of the synthetic rubber neoprene which has the property of very high elastic extensibility. Certain non-spherical modes, which occur between the maximum and minimum pressures, have been noted by Alexander, but here we shall assume that the balloon remains spherical throughout the deformation.

Provided that the thickness of the skin of the balloon is very much less than the radius, the deformation in the skin may be regarded as corresponding to a uniform equibiaxial tension or, equivalently, unidirectional compression. We assume that the undeformed thickness of the skin is equal to h . The incompressibility assumption implies that the actual volume, V say, of the rubber constituting the balloon remains constant during inflation. Let r be the radius when the pressure is P and assume $h \ll r$. When a is the extension ratio in the surface of the balloon the thickness of the skin is ha^{-2} . In the strained state the volume of material is therefore approximately $4\pi r^2 ha^{-2} (= V)$ so we may write

$$ra^{-1} = \left(\frac{V}{4\pi h} \right)^{\frac{1}{2}} = r_0, \quad (20)$$

where r_0 is the unstrained radius. Let T be the surface tension in the skin so that, by equation (18), $P = 2Tr^{-1}$. The principal stress σ in the surface is given by

$$\sigma ha^{-2} = T, \quad (21)$$

and equation (17) is applicable. By use of equations (17), (18), (20) and (21) we can therefore express the pressure in terms of a as

$$Pr_0/2h = \sigma a^{-3} = \mu_r[a^{\alpha_r-3} - a^{-(2\alpha_r+3)}]. \quad (22)$$

With the particular values of the parameters α_r and μ_r appropriate to the strain-energy function constructed in §4 the curve of $Pr_0/2h$ against a is plotted in figure 5. This curve is, in effect, equivalent to the equibiaxial tension curve shown in figure 4, the ordinate being reduced by the factor a^{-2} .

As pointed out in §4 the values of the parameters used here relate specifically to the rubber used in Treloar's experiments and one would therefore not expect the curve in figure 5 to correspond exactly with the data from Alexander's experiments on neoprene. Alexander's (1971) paper does not contain numerical data sufficient to

† My thanks are due to Dr R. Hill, F.R.S., who recommended a calculation of this effect.

determine values of the parameters accurately for neoprene, so such values cannot be given here. However, the predictions of the curve are qualitatively correct as regards neoprene. Our theory predicts that the maximum pressure occurs when $a^3 \simeq 2.2$, i.e. when the volume enclosed by the balloon has slightly more than doubled, and that the minimum occurs for $a \simeq 4$, i.e. when the volume increase is about 64 times. These figures are corroborated by the experiments of Alexander (1971) and Hart-Smith (1966) on balloons and by those of Rivlin & Saunders (1951) on circular membranes. An account of this problem in relation to the Mooney form of strain energy is given by Green & Adkins (1960, §4.13).

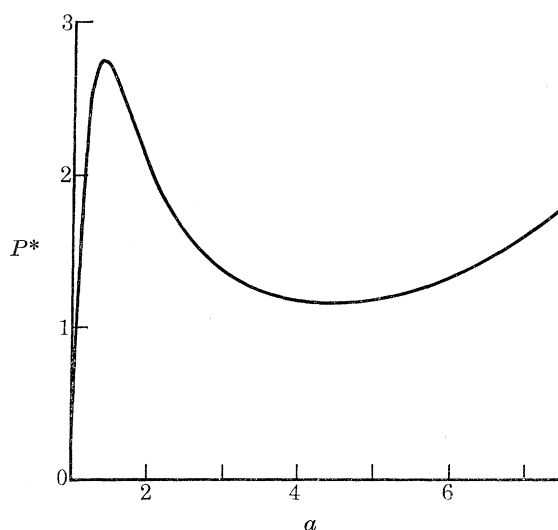


FIGURE 5. Pressure of inflation of a spherical balloon according to the three-term theory.

It is instructive to determine analytically the maximum pressure for a single-term strain-energy function $\mu\phi(\alpha)$. Setting $r_0 P/2h = P^*$ and differentiating P^* with respect to a we obtain†

$$a^4 \frac{dP^*}{da} = \mu[(\alpha - 3)a^\alpha + (2\alpha + 3)a^{-2\alpha}].$$

Equating dP^*/da to zero yields

$$a^{3\alpha} = \frac{2\alpha + 3}{3 - \alpha},$$

so that a positive, real maximum exists only for values of α within the range $-\frac{3}{2} < \alpha < 3$. The single-term strain-energy functions that have been discussed in §4 all correspond to values of α which lie within this range and therefore all predict some maximum pressure. However, for no value of α within this range does a minimum pressure exist: for such values the pressure reaches a maximum and then

† $P > 0$ for all $a > 1$ if and only if $\mu\alpha > 0$.

decreases eventually to zero. For values of α outside this range P^* increases indefinitely and there is neither a maximum nor a minimum. Such situations are physically unrealistic and, clearly, no single-term strain energy of the type considered is sufficient to describe all the phenomena discussed here.

6. CONSTITUTIVE INEQUALITIES

A problem intimately related to that of determining forms of the strain-energy function is the one propounded by Truesdell (1956). He considered that the most important unsolved problem in the theory of finite deformation elasticity was that of determining the restrictions which are to be placed on the strain-energy function to ensure physically reasonable response. This problem has received much attention but only recently has an adequate solution been found. Hill (1968*a, b*) proposed a class of inequalities for independent and critical examination in the light of available experimental data for elastic and plastic solids, and subsequently (1970*b*) he considered these inequalities in the context of isotropic elasticity. He has shown that one and only one of the inequalities (henceforth referred to as Hill's inequality) admits incompressibility and, furthermore, has indicated that this same inequality is also the best for compressible solids. Ogden (1970*a, b*), on the basis of compressibility data for continuum and foam rubbers, has demonstrated that Hill's inequality is the only member of the class which is entirely consistent with this data.

Here we briefly consider the analysis of this inequality in relation to the strain-energy function (13). For incompressible solids Hill's inequality asserts that the scalar product $(\mathcal{D}\boldsymbol{\sigma}/\mathcal{D}t)\boldsymbol{\epsilon}$ is positive definite at all strains and for arbitrary non-zero strain-rates. Symbolic notation is used here, $\boldsymbol{\epsilon}$ representing the Eulerian strain-rate and $\boldsymbol{\sigma}$ the Cauchy stress. The notation $\mathcal{D}/\mathcal{D}t$ is used for the rigid-body derivative (the rate of change on axes rotating rigidly with the local body spin).

It may readily be shown (see equation (A. 10) of the appendix) that, relative to the principal axes of the Eulerian strain ellipsoid (also the principal axes of Cauchy stress in the case of isotropic elasticity), the body spin has non-zero components ω_{ij} ($i \neq j$), on those axes, given by

$$-\omega_{ij} = \frac{a_i^2 + a_j^2}{a_i^2 - a_j^2} \epsilon_{ij}, \quad (23)$$

where ϵ_{ij} are specifically the components of the Eulerian strain-rate on the same axes. We assume in (23) that $a_i \neq a_j$ for $i \neq j$; in situations where $a_i = a_j$ for $i \neq j$ a limiting procedure can be used at the appropriate place. From equation (A. 8) of the appendix, we also have

$$\dot{a}_i/a_i = \epsilon_{ii} \quad (i = 1, 2, 3), \quad (24)$$

where the superposed dot denotes the material time derivative.

Let $\hat{\sigma}_{ij}$ denote the components of the rigid-body derivative of Cauchy stress on the axes of the Eulerian strain ellipsoid. Then, by use of equations (14), (23) and (24), we obtain

$$\hat{\sigma}_{ii} \equiv \dot{\sigma}_i = \mu_r \alpha_r a_i^{\alpha_r} \epsilon_{ii} - \dot{p} \quad (i = 1, 2, 3),$$

with summation over r , and

$$\hat{\sigma}_{ij} \equiv -\omega_{ij}(\sigma_i - \sigma_j) = \mu_r \frac{a_i^2 + a_j^2}{a_i^2 - a_j^2} (a_i^{\alpha_r} - a_j^{\alpha_r}) \epsilon_{ij} \quad (i \neq j),$$

again with summation over r . It follows that the scalar product $(\mathcal{D}\boldsymbol{\sigma}/\mathcal{D}t)\boldsymbol{\epsilon}$ is expressible as

$$\hat{\sigma}_{ij} \epsilon_{ij} \equiv \mu_r \alpha_r a_i^{\alpha_r} \epsilon_{ii}^2 + \mu_r \frac{a_i^2 + a_j^2}{a_i^2 - a_j^2} (a_i^{\alpha_r} - a_j^{\alpha_r}) \epsilon_{ij}^2. \quad (25)$$

In the first group of terms on the right-hand side of (25) there is summation over i and r , while in the second group there is summation over r , and over i and j ($i \neq j$). The indices i and j range from 1 to 3 while r ranges over the number of terms included in the strain-energy function (13). Note that, by virtue of the identity $\epsilon_{11} + \epsilon_{22} + \epsilon_{33} = 0$, the arbitrary hydrostatic pressure p is absent from the expression (25).

In general necessary and sufficient conditions on the parameters α_r and μ_r that ensure the expression (25) is positive definite do not exist. However, the conditions

$$\mu_r \alpha_r > 0 \quad (\text{each } r; \text{ no summation}) \quad (26)$$

are clearly sufficient. When the strain-energy function comprises either one or two (α, μ) terms only these conditions are also necessary. When there are three or more (α, μ) terms not all the conditions (26) are necessary. The inequalities $\mu_r \alpha_r > 0$ are plainly satisfied for the particular (α, μ) pairs constructed in §4 and, so, our strain-energy function is fully consistent with Hill's inequality. Moreover, Hill's inequality is entirely consistent with the known behaviour of rubberlike solids as viewed in terms of the present strain-energy function. This provides further evidence in favour of the general adoption of Hill's inequality as a basis for delimiting other possible forms of strain-energy function.

APPENDIX. SOME KINEMATICAL RESULTS

We denote the deformation gradient by A ; by the polar decomposition theorem we may write

$$A = RU = VR, \quad (\text{A } 1)$$

where U and V are positive definite symmetric matrices and R is a proper orthogonal matrix. The right stretch U is coaxial with the Lagrangian strain ellipsoid and the left stretch V with the Eulerian strain ellipsoid. Taking the material derivative of A we obtain

$$\dot{A} = \Gamma A = \dot{R}U + R\dot{U}, \quad (\text{A } 2)$$

where Γ is the matrix of velocity gradients. The Eulerian strain-rate, denoted by Σ , is given by

$$\Sigma = \frac{1}{2}(\Gamma + \Gamma') = \frac{1}{2}R(\dot{U}U^{-1} + U^{-1}\dot{U})R', \quad (\text{A } 3)$$

where a prime denotes the transposed matrix, and the body spin, denoted by Ω , is given by

$$\Omega = \frac{1}{2}(\Gamma - \Gamma') = \dot{R}R' + \frac{1}{2}R(\dot{U}U^{-1} - U^{-1}\dot{U})R'. \quad (\text{A } 4)$$

The Green–St Venant strain measure E is defined by

$$E = \frac{1}{2}(A'A - I) \quad (\text{A } 5)$$

and has principal values e_i given by

$$e_i = \frac{1}{2}(a_i^2 - 1) \quad (i = 1, 2, 3). \quad (\text{A } 6)$$

Taking the material derivative of E and making use of equation (A 2)₁, we obtain

$$\dot{E} = A'\Sigma A. \quad (\text{A } 7)$$

Let T denote the matrix of any symmetric second-order tensor with principal values t_1, t_2 and t_3 . Suppose the ‘spin’ of the principal axes of T has components ω_{ij} on those axes. It follows that the material derivative of T has normal components $\dot{t}_i$ ($i = 1, 2, 3$) and shear components $-\omega_{ij}(t_i - t_j)$ ($i \neq j$) when decomposed on the principal axes of T . Applying these results to E , so that ω_{ij} represents the spin of the Lagrangian strain ellipsoid, and decomposing equation (A 7) on the axes of the Lagrangian strain ellipsoid, we obtain

$$\dot{e}_i = a_i^2 \epsilon_{ii} \quad (i = 1, 2, 3)$$

and

$$-\omega_{ij}(e_i - e_j) = a_i a_j \epsilon_{ij} \quad (i \neq j),$$

where ϵ_{ij} are the components of Σ on the axes of the Eulerian strain ellipsoid. On using equations (A 5) there follow the relations

$$\dot{a}_i/a_i = \epsilon_{ii} \quad (i = 1, 2, 3) \quad (\text{A } 8)$$

and

$$-\omega_{ij} = \frac{2a_i a_j \epsilon_{ij}}{a_i^2 - a_j^2} \quad (i \neq j), \quad (\text{A } 9)$$

assuming $a_i \neq a_j$ for $i \neq j$. The formulae (A 9) give the components of spin of the Lagrangian ellipsoid on its own axes in terms of the shear components of the Eulerian strain-rate on the axes of the Eulerian strain ellipsoid.

Next, noting that U has normal components $\dot{a}_i$ ($i = 1, 2, 3$) and shear components $-\omega_{ij}(a_i - a_j)$, $i \neq j$, on the principal axes of U and using (A 8) and (A 9) in equation (A 4), we obtain the components ω_{ij}^E of the spin of the Eulerian strain ellipsoid on its own axes. A short calculation leads to

$$-\omega_{ij}^E = \frac{a_i^2 + a_j^2}{a_i^2 - a_j^2} \epsilon_{ij} + \text{the body spin} \quad (i \neq j). \quad (\text{A } 10)$$

A derivation of the formulae (A 9) and (A 10), based on elementary geometrical arguments, was first given by Hill (1970*a, b*). However, since these formulae are not well known, it was thought worthwhile to give another derivation. The method used here was adopted by Ogden (1970*b*).

The author is grateful to Dr R. Hill, F.R.S., for suggesting this investigation, to Professor L. R. G. Treloar for supplying numerical data relating to his experimental work, and to Professor P. Chadwick for helpful discussions.

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